DAY 03 · THEORY HANDOUT

Pandas: Read and Clean Tables

We follow visible table changes, from the first DataFrame to filtering and grouped summaries.

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A short introduction to NumPy

NumPy is a Python library for arrays: groups of values stored in a regular layout.

Python list

[40, 60, 80]

A general collection.

NumPy array

[40, 60, 80]

Works on number groups.

Pandas table

name | score

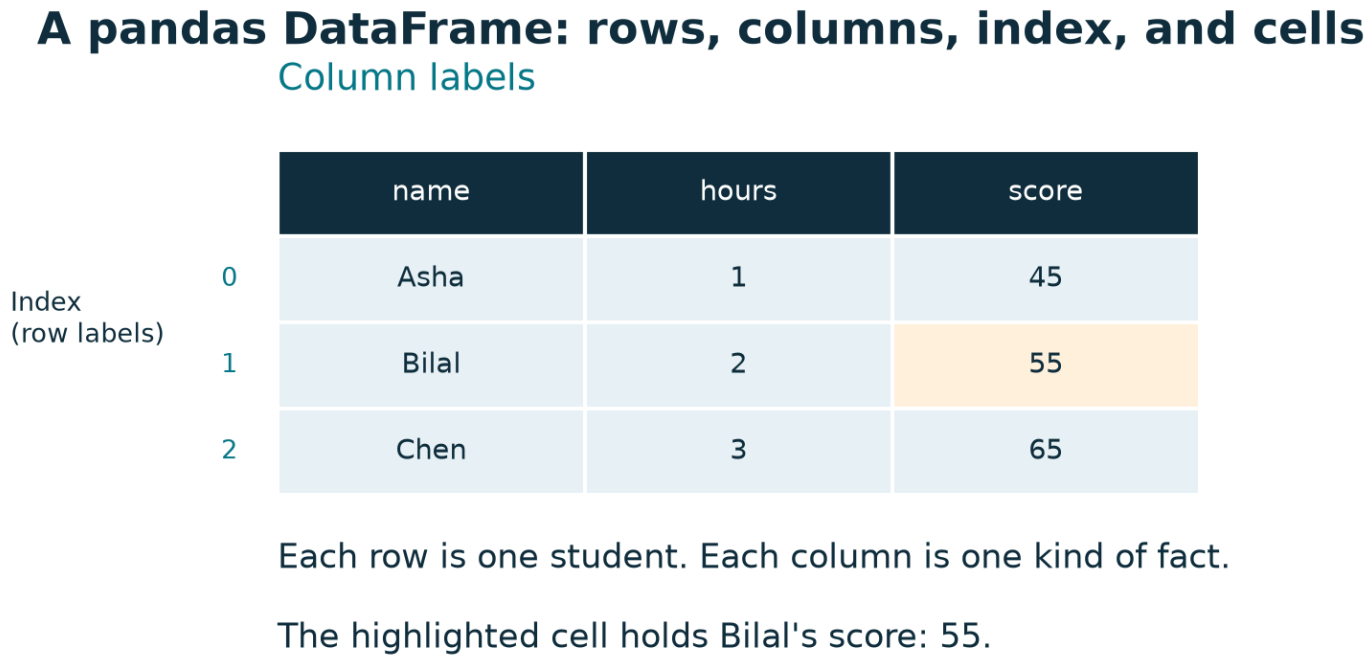
Labels rows and columns.

A one-dimensional array is a row of values. A two-dimensional array is a grid. Adding 5 to a NumPy array gives **[45, 65, 85]** from **[40, 60, 80]**.

Pandas can use NumPy arrays and adds table labels and data tools. NumPy is also used by many ML libraries. We keep it installed for these tools.

> In these lessons, we mainly use named Pandas columns. An array is still useful when a model needs groups of numbers.

DataFrame anatomy



Each row is one student. Each column holds one kind of information.

Definition

A **DataFrame** is a two-dimensional labeled table. A **cell** is the value at the meeting of a row and column.

How it works

The **index** contains row labels. The **column labels** name the information. A name can be text while a score is a number.

> Find Bilal's score. Use his row and the score column: 55.

Series and DataFrame

Series versus DataFrame

	score
0	45
1	55
2	65

df['score']
Series: one dimension

	name	score
0	Asha	45
1	Bilal	55
2	Chen	65

df[['name', 'score']]
DataFrame: two dimensions

The outer brackets select. The inner list says we want a DataFrame with these columns.

Definition

A **Series** is one-dimensional labeled data. One selected column is normally a Series.

How it works

df["score"] gives a Series. df[["score"]] gives a one-column DataFrame. df[["name", "score"]] gives a two-column DataFrame.

A one-column DataFrame still has two dimensions: rows and columns.

Inspecting a table

Tool	What it shows	Why we use it
head(3)	First three rows	See example records
shape	Row count and column count	Check table size
dtypes	Type of each column	Check text versus numbers
info()	Types and non-missing counts	Spot gaps
describe()	Numeric summary statistics	Check centre and range

Attributes such as shape have no parentheses. Methods such as head() are called with parentheses.

Definition

Inspection means looking at the data before changing it. A wrong type or missing value can affect later calculations.

How it works

In describe(), count excludes missing values. The 50% row is the median. A numeric student ID is a label; its average usually has no useful meaning.

> Check what each column means before deciding which statistics to calculate.

loc and iloc

loc uses labels; iloc uses positions

	name	group	hours	practice	score
101	Asha	A	1	2	45
102	Bilal	B	2	4	55
103	Chen	A	3	5	65

by_id.loc[102, 'score'] = 55
row label 102, column label 'score'

by_id.iloc[1, 4] = 55
row position 1, column position 4

The row label 102 is different from row position 1. Both shown selections return 55.

Definition

loc selects using labels. **iloc** selects using integer positions.

How it works

Positions start at 0. A loc label slice includes the stop label. An iloc position slice excludes the stop position.

> by_id.loc[102, "score"] and by_id.iloc[1, 4] select the same cell here.

Boolean filtering

pandas: check both rules, then keep the matching rows

Name	hours	group	hours >= 3	group == 'A'	Both &
Asha	1	A	False	True	-
Bilal	2	B	False	False	-
Chen	3	A	True	True	KEEP
Diya	4	B	True	False	-
Eshan	2	A	False	True	-
Fatima	5	B	True	False	-
Gita	3	A	True	True	KEEP
Hari	4	B	True	False	-

Result: Chen and Gita. The mask has one answer for every row.

Chen and Gita meet both rules. Diya, Fatima, and Hari meet the hours rule but are not in group A.

Definition

Filtering keeps records that satisfy a condition. A **Boolean mask** gives one True or False result per row.

How it works

The first rule checks hours >= 3. The second checks group == "A". & combines both masks. students[mask] keeps only Chen and Gita.

> Diya meets the hours rule but not the group rule, so her row is left out.

Membership with isin()

isin() asks whether each name is in the allowed list

Name	isin(['Asha', 'Diya'])	Selected
Asha	True	Yes
Bilal	False	No
Chen	False	No
Diya	True	Yes

isin() returns a Boolean Series. Use it inside students[...] to keep rows.

Asha and Diya appear in the allowed list, so their mask values are True.

Definition

isin() checks whether each value appears in a chosen list. It returns a True or False answer for each row.

How it works

The allowed names are Asha and Diya. Each name is checked against that list. The outer students[...] keeps the True rows.

> isin() compares whole values. It does not search for a short piece inside a name.

Missing values

fillna(): replace an empty value and keep known scores

Name	Original score	score_filled
Asha	45	45
Gita	NaN	65.57
Hari	74	74

Known scores total 459. There are 7 known scores.

Mean = 459 / 7 = 65.57. Only the missing value is filled.

fillna(mean_score) fills only missing scores. The original score column stays available for comparison.

Definition

A **missing value** means a value is unavailable. Missing is different from zero. NaN is one way pandas represents a missing number.

How it works

Seven known scores total 459. Their mean is 459 / 7, about 65.57. fillna(mean_score) replaces only the missing score. The new column keeps the old data available.

> A filled mean is an estimate, not a real score. Keep the original data available.

GroupBy and aggregation

groupby(): split rows, calculate, then collect the answers

Group A hours: 1, 3, 2, 3
Mean: 9 / 4 = 2.25

Group B hours: 2, 4, 5, 4
Mean: 15 / 4 = 3.75

Group	Mean hours
A	2.25
B	3.75

```
students.groupby('group')['hours'].mean()
```

Rows are split by group. Only hours are selected. mean calculates one answer for each group.

Definition

groupby() groups rows by a shared key. **Aggregation** reduces values to a summary such as count, mean, or maximum.

How it works

students.groupby("group")["hours"].mean() means: split by group, select hours, and find each group mean.

Group A: 9 / 4 = 2.25 hours. Group B: 15 / 4 = 3.75 hours.

Merging related tables

merge(): attach a room using the matching group label

Name	Group
Asha	A
Bilal	B
Chen	A

Group	Room
A	Room 1
B	Room 2

match group →

Name	Room
Asha	Room 1
Bilal	Room 2
Chen	Room 1

on='group' chooses the shared key. how='left' keeps every student row.

Each student finds the room with the same group label. A left join keeps the student even if no room matches.

Definition

A **merge** combines records using a shared key. It is useful when related information lives in two tables.

How it works

students.merge(rooms, on="group", how="left") keeps the student rows and adds matching room information. Repeated keys on both sides can create extra rows, so inspect the result.

A merge matches keys. concat() joins tables along an axis; it does not perform the same key match.

DAY 03 · COMPLETE

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